

Emir Ceyani

PhD Student in Electrical & Computer Engineering, University of Southern California

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Research Interests

- Large Language Models
- Graph Neural Networks
- Generative Adversarial Networks
- Flow-Based Generative Models
- Conformal Prediction
- Uncertainty Quantification
- Federated Learning
- Bayesian Methods
- AI For Science & Engineering

Education

University of Southern California, PhD in Electrical & Computer Engineering **Jan 2021 – May 2026**
Thesis: Federated Learning over Graphs. Advisor: Salman Avestimehr *Los Angeles, California*

Bilkent University, MSc in Electrical Engineering **Jun 2018 – Dec 2020**
Thesis: Spatiotemporal Forecasting over Graphs with Deep Learning *Ankara, Türkiye*

Bilkent University, BSc in Electrical Engineering **Sep 2013 – Jun 2018**

Research Experience

University of Southern California, Graduate Research Assistant **Jan 2021 – Present**

- **AI-Based Analog & RF Circuit Design:** Designing graph neural networks and transformers to automate analog and RF circuit design process including topology selection, layout optimization, and AI-guided circuit design.
- **Conformal Prediction for LLMs:** Developing uncertainty-aware LLM fine-tuning methods based on conformal prediction for close-ended and open-ended tasks to reduce the need of calibration data and extra post-hoc calibration step meanwhile improving learned representations.
- **Theory & Applications of Generative Flow Networks:** Researching on the fundamentals and the applications of generative flow networks (GFlowNets) including sample efficiency, exploitation-exploration tradeoffs, and applications to hyperbolic geometry and federated learning (*SIAM-SDM'25*) In collaboration with Prof. Salem Lahlou from MBZUAI.
- **Federated MRI Image Generation:** Designed the personalized & federated image generation framework using generative adversarial networks for the first time in the literature (*MIA'24, IEEE TMI'23, ISBI'23, ISMRM'23*.)
- **Federated Graph Learning:** Proposed the problem of training of graph neural networks in a federated setting and released the first benchmarking study to the literature. Improved personalization of federated graph neural networks (*SIAM-SDM'25 & AAAI'22*.)

TensorOpera AI (formerly FedML AI), Research Scientist Intern **May 2022 – Aug 2022**

- **FedGraphNN:** Developed a secure & private data sharing platform for GNNs tailored for recommendation systems and drug discovery.

Turkcell Technologies, Research & Development Engineer **Oct 2018 - Dec 2020**

- **Spatiotemporal Forecasting with ConvLSTMs:** Developed and deployed a spatio-temporal forecasting algorithm with convolutional LSTM models.

Publications

Published and Accepted Journal Publications

- [1] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, “One model to unite them all: Personalized federated learning of multi-contrast MRI synthesis.” In *Medical Image Analysis*, Volume 94, May 2024, 103121.
- [2] G. Elmas, S. UH. Dar, Y. Korkmaz, **E. Ceyani**, B. Susam, M. Ozbey, S. Avestimehr, T. Cukur. “Federated Learning of Generative Image Priors for MRI Reconstruction.” in *IEEE Transactions on Medical Imaging*, July 2023.

Publications in Conference Proceedings & Workshops (Peer Reviewed)

- [1] **E. Ceyani**, H. Xie, B. Buyukates, C. Yang, and S. Avestimehr, “FedGrAINS: Personalized Subgraph Federated Learning with Adaptive Neighbor Sampling.” In *SIAM International Conference on Data Mining (SDM25)*, 2025. (acceptance rate: 26.7%)
- [2] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, “A Personalized Federated Learning Approach for Multi-Contrast MRI Translation,” in 31st annual meeting of International Society for Magnetic Resonance Imaging (ISMRM), Toronto, Canada, June 2023.
- [3] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, “Personalized, Federated, And Unified MRI Contrast Synthesis,” in IEEE 20th International Symposium on Biomedical Imaging (ISBI), Virtual Conference, Apr. 2023. (Presented online)
- [4] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, “pFLSynth: Personalized Federated Learning of Image Synthesis in Multi-Contrast MRI,” in NeurIPS Medical Imaging Meets, Virtual Conference (oral), Dec. 2022. (Presented online)
- [5] C. He*, **E. Ceyani***, K. Balasubramanian*, M. Annavaram, and S. Avestimehr, “SpreadGNN: Serverless Multi-task Federated Learning for Graph Neural Networks,” accepted to *AAAI*, 2022. (acceptance rate: 15%) (co-first)
- [6] C. He*, K. Balasubramanian*, **E. Ceyani***, C. Yang, H. Xie, L. Sun, L. He, L. Yang, P. S. Yu, Y. Rong, P. Zhao, J. Huang, M. Annavaram, and S. Avestimehr, “FedGraphNN: A Federated Learning System and Benchmark for Graph Neural Networks,” in DPML workshop at ICLR & GNNsSys workshop at MLSys, 2021, co-first
- [7] T. Ergen, and **E. Ceyani**. “A highly efficient recurrent neural network architecture for data regression.” 2018 26th Signal Processing and Communications Applications Conference (SIU). IEEE, 2018.

Professional Service

- **Reviewer:** ICLR’24, NeurIPS’22-’23 (Main & Dataset and Benchmark Tracks), Federated Learning with Graph Data (FedGraph2022-2023), CrossFL-MLSYS’22, IEEE Transactions of Neural Networks and Learning Systems, IEEE Transactions of Big Data
- **Organizer & Technical Committee Member:** 1st & 2nd International Workshops on Federated Learning with Graph Data, Technical Program Committee Member at the CrossFL Workshop, MLSYS2022
- **Summer Schools Attended:** LOGML’22 Summer School, ProbAI 2021 Summer School, PAISS Summer School’19, Deep—Bayes’19 Summer School
- **Teaching Assistant:** ProbAI 2021 Summer School

Technical Skills

Programming Languages: Python, Matlab, Java, C++, Julia

Developer Tools: L^AT_EX, VS Code, Jupyter, Inkscape

Technologies/Frameworks: Linux, PyTorch, PyTorch Geometric, TensorFlow, NumPy, Matplotlib, Pandas, Github

Hobbies

Playing Bass, 8-ball pool, Trading card games, Martial arts, Cooking